MIL-A-35045D
27 April 1988
SUPERSEDING
MIL-A-35045C
13 December 1978

MILITARY SPECIFICATION

APPLESAUCE, INSTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers (dehydrated) instant applesauce for use by the Armed Forces as an item of general issue and as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-96	- Cans, Metal, 28 Gauge and Lighter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8915

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

United States Standards for Condition of Food Containers

(Copies may be obtained from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0610, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, SW, Washington, DC 20460.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Copies should be obtained from the National Academy of Sciences, National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN DAIRY PRODUCTS INSTITUTE

Standards for Grades of Dry Milks, Including Methods of Analysis, Bulletin 916

(Copies should be obtained from the American Dairy Products Institute, 130 N. Franklin St., Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Copies should be obtained from the American Trucking Associations, Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Copies should be obtained from the Uniform Classification Committee, Suite 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Apples. Apples shall be properly matured, sound, fresh, clean apples of the varieties commonly used for preparing applesauce. Dried (or evaporated) apples of the latest season's crop of fresh apples that meet the above requirements and that have been properly prepared by washing, sorting, trimming, peeling, coring, cutting, blanching or cooking, sulfiting, and drying may be substituted for fresh apples. The dried apples shall possess a clean, typical dried apple aroma and flavor, free from any objectionable flavor or odor other than that due to a proper treatment with sulfite.

3.2.2 Water. Water used for cooking, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Sweetening ingredients. Sweetening ingredients shall be sugar (sucrose), liquid sugar, invert sugar syrup, any combination thereof, or a mixture of one or more of the aforementioned ingredients with dextrose or corn syrup (liquid or dry), or both, in such proportions as will result in a finished product complying with 3.5.

3.2.4 Sulfites. Sulfites shall meet the requirements of the Food Chemicals Codex.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

3.2.5 Anticaking agents. Anticaking agents shall be white, impalpable powders that are of such particle size that at least 90 percent shall pass through a U.S. Standard No. 325 sieve when tested in accordance with 4.5.7.1 and of such quality as to prevent caking in applesauce without affecting its flavor.

3.2.6 Optional ingredients. One or more of the following ingredients may be added to the applesauce either before or after dehydration:

Lemon juice (single strength, concentrated or dehydrated). If not dehydrated, addition shall be such as to prevent wet areas and caking.

The following shall meet the requirements of the Food Chemicals Codex:

Malic acid
Citric acid
Salt (sodium chloride). Iodized salt shall not be used.
Sodium citrate

3.3 Processing. The apples shall be cleaned, washed, and processed for removal of peel, seed, stem, carpel tissue, blossom, end, and defective and other objectionable material. The prepared apples shall be blanched or cooked to such extent as necessary to comply with 3.5.2 and shall be treated with sulfite in such quantities as necessary to comply with 3.5.1 prior to rehydration. After sulfiting, contact of the wet apples with iron shall be strictly avoided. The sulfited and blanched or cooked apples shall be dehydrated either with or without the sweetening and optional ingredients to meet the requirements of 3.5. 1/ An anticaking agent shall be added meeting the requirements in 3.5.1 and meeting FDA guidelines.

1/ If dried apples are substituted for fresh apples, they shall be sulfite treated and shall meet the same requirements as fresh apples.

3.4 Can filling. Each can shall be filled with 22 ounces of product (see 5.1.1).

3.5 Finished product requirements.

3.5.1 Dehydrated product. The dehydrated product shall possess a good typical apple aroma, free from hay-like, scorched, musty, or other objectionable odors, and shall be of such particle size that 100 percent, by weight, shall pass through a U.S. Standard No. 12 sieve. Anticaking agent shall be added in quantities sufficient to prevent caking without affecting flavor, when examined in accordance with 4.5.7. If the product is rejected because of caking as per table I, test for caking need not be performed. Lumps that are fragile in nature and that are readily broken by shaking or light pressure shall not be considered as caking. The sulfite content shall be 350 ± 150 parts per million (calculated as sulfur dioxide as tested in 4.5.7). The average moisture content of the dehydrated product shall not

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

exceed 2.5 percent by weight, when tested as specified in 4.5.7. The product shall be free from foreign material such as splinters, paint, varnish, rubber, metal, and other objectionable foreign matter and from evidence of insect infestation.

3.5.2 Rehydrated product. The rehydrated applesauce shall possess a distinct, pleasing, characteristic apple flavor free from flavors due to overripe apples, oxidation, fermentation or carmelization, or ground flavors due to storage or other causes or any other undesirable flavor and shall be within a pH range of 3.0 to 4.2 (when tested as specified in 4.5.7) and the degrees Brix range (when tested as specified in 4.5.7) applicable to the specific pH as follows:

pH	Degree Brix range
3.0	24.5 to 25.5
3.1	24.0 to 25.5
3.2	21.0 to 25.5
3.3	18.5 to 25.5
3.4	17.0 to 24.5
3.5	16.5 to 21.5
3.6	16.5 to 20.5
3.7	16.5 to 18.5
3.8	16.5 to 18.0
3.9	16.5 to 17.0
4.0	16.5 to 17.0
4.1	16.5 to 16.7
4.2	16.5 to 16.6

The rehydrated applesauce shall be such that the apple particles are evenly divided, granular, fairly tender, but not hard or mushy, not lumpy and not more than slightly "salvy" or "pasty." The rehydrated applesauce shall have a viscosity such that the product does not flow more than 8.5 centimeters and shall not have a liquid separation of more than 1.0 centimeter when tested in accordance with footnote 5/ to table I. The rehydrated applesauce shall be reasonably free from defects. Reasonably free from defects means that seeds or any portion thereof, peel not blending with the applesauce color or tough even though not visually noticeable and discolored apple particles do not exceed the area of 1 square centimeter, and no more than four stamen (hairlike substances from the blossom end of the apple) are present in the product when tested in accordance with footnote 5/ to table I and any carpel tissue that may be present is no more than slightly noticeable upon eating the product.

3.5.3 Palatability. The product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualifications. The product and ingredients shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21 CFR Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection service necessary to assure compliance with the requirements of this document.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to any identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

4.5.1.2 Testing of cans. Cans shall be tested for tin plate in accordance with PPP-C-96. The lot size shall be expressed in units of cans with lids. The sample unit shall be one can with lid. The sample size shall be the number of sample units indicated by inspection level S-1. Results shall be reported to the nearest 0.01 lb/base box. An average of less than 0.25-pound per base box commercial electrolytic tin plate is a test failure. Tests on can label and external can coating (when requested) shall be performed on the same cans submitted for tin plate testing, in accordance with the pertinent subsidiary document. Any test failure shall indicate unacceptable cans and use of such shall be cause for rejection of the involved end item.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.2.1 Fresh apple examination. When fresh apples are utilized, examination shall be performed immediately before processing of the product to determine conformance to 3.2. Nonconformance shall be cause for rejection of the involved finished product.

4.5.3 Net weight inspection. The net weight of the filled and sealed cans shall be determined by weighing each sample unit on a suitable scale tared with a representative empty can. Results shall be reported to the nearest 1/4 ounce. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for minor defects. Any individual net weight of less than 21-1/2 ounces shall be scored as a minor defect. If the average net weight is less than 22 ounces, it is a major defect and the lot shall be rejected.

4.5.4 Product inspection. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in cans. The sample unit shall be the contents of one filled and sealed can. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	
	Dehydrated product
101	Hay-like, scorched, musty, or other objectionable odors
102	Caked, or presence of lumps that do not break up with shaking or light finger pressure
103	Less than 100 percent passes through a U.S. Standard No. 12 sieve 3/
	Rehydrated product 4/
104	Not a pleasing distinct characteristic flavor free from undesirable flavors
105	Carpel tissue more than slightly noticeable during eating
106	Quality of apple particles not as specified (see 3.5.2)
107	Seed particles, noticeable peel particles, and discolored apple particles exceed an area of 1 square centimeter 5/
108	Not good typical apple aroma
109	More than four stamen 5/
110	Viscosity flow more than 8.5 centimeters 5/
111	Liquid separation more than 1.0 centimeter 5/

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal) or foreign odor or flavor (e.g., burnt, scorched, stale, sour, rancid, musty, carmelized) shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (The comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ One hundred grams of the product shall be screened through a U.S. Standard No. 12 sieve (0.066-inch openings) equipped with a tight-fitting cover and catch pan. Hold the sleeve assembly in the left hand at an angle of approximately 30 degrees from the horizontal, and strike the sieve assembly for 2 minutes with short sharp strokes of the right hand, at the rate of 150 times per minute. After each 25 strokes, tip the sieve assembly approximately 60 degrees so that the sieve assembly is now tilted approximately 30 degrees from the horizontal in the opposite direction. Care shall be exercised to keep moisture absorption by the product to a minimum. Make one determination per sample unit.

4/ The instant applesauce shall be rehydrated by adding 5.5 avoirdupois ounces of the product to 24 fluid ounces of cold water ($55^{\circ} \pm 5^{\circ}F$), mixing thoroughly, and allowing to stand for 10 minutes.

5/ For viscosity, liquid separation, and presence of defects, the following procedure shall be followed: Use a 16-inch square sheet of paper containing concentric circles 0.5 cm apart that is laminated in plastic for a flow sheet. Lay the flow sheet on a flat surface under good lighting. Mix a rehydrated applesauce sample thoroughly at approximately room temperature and transfer it to a 600 mL beaker or No. 2-1/2 can. Align a plastic cylinder (inside diameter of 3 inches and length of 3-1/4 inches) with the periphery of the center circle. While holding the cylinder firmly in place, pour applesauce from the beaker into the cylinder till full with a slight excess. Use a straight edge to level off the top of the cylinder, taking care to avoid spilling any applesauce on the flow sheet and disregarding any free liquid at this time. Lift the cylinder perpendicularly at least 3-1/2 inches with a smooth, even motion. Determine the extent of spread after 1 minute by averaging the readings at four quadrants of the flow sheet. Readings shall be taken at the edge of the applesauce, including free liquid. Liquid separation is determined by averaging measurements taken at the four quadrants. To determine the presence of defects, use a 16-inch plastic spreader with a 3/16 inch notch 15 inches along one edge to further spread the applesauce into a uniform layer, taking care to hold the spreader in a vertical position with a minimum of tilting to maintain a constant depth of applesauce. Pick out all seed particles, noticeable peel, discolored apple particles, and stamen, and place them, exclusive of stamen, in a contingent position on a square centimeter area. Count the stamen separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

4.5.5 Testing for oxygen content of cans. The cans shall be tested for oxygen content in accordance with American Dry Milk Institute Bulletin 916, Standards for Grades of Dry Milk, Chapter: Methods of Analysis; Determination of Oxygen. When the can has been sealed under reduced pressure, the gases may have to be extracted and compressed before testing for oxygen. The lot size shall be expressed in units of filled and sealed cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 percent. Residual oxygen content of more than 2.0 percent in a can shall be a major defect.

4.5.6 Can leakage testing. The filled and sealed cans shall be tested for leakage by submerging in water contained in a desiccator or other suitable container and maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) for 30 seconds and observe for leakage. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as signs of leakage. The lot size shall be expressed in units of filled and sealed cans. The sample unit shall be one filled and sealed can. Leakage in a can shall be a major defect. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.7 Finished product testing. The finished product shall be tested for caking, sulfur dioxide, moisture, pH, and Brix. (Conformance to pH-Brix relationship requirements shall be ascertained from the results of the pH and Brix tests.) The lot size shall be expressed in units of filled and sealed cans. The sample unit shall be a 1-pound composite derived from the sample size. The inspection level shall be S-2. Each result shall be reported to the same unit or decimal place as specified for the pertinent requirement except that the caking test shall be reported as pass or fail. Test for Brix and pH shall be performed on the rehydrated product prepared in accordance with footnote 4/ of table I. A test failure shall be a major defect and cause for rejection of the lot. If the cans have been sealed under the alternative gas packing procedure outlined in 5.1.1, the oxygen content need not be determined. If the cans have not been sealed under the alternative procedure, the samples shall be tested for residual oxygen content requirement and the inspection and rejection criteria thereto applied.

4.5.7.1 Product caking. Test tubes that are 25mm in outside diameter and that are fitted with plastic screw tops with inert liners shall be preheated in a water bath at 120° ± 2°F for at least an hour. Approximately 25 grams of dehydrated applesauce shall be placed in the test tubes. Caps shall be tightly screwed onto the test tubes, and the filled test tubes shall be immersed in the water bath at 120°F for 1 hour. After 1 hour, tubes containing dried applesauce shall be examined for caking. Caked applesauce that breaks apart upon slight tapping of the test tube on the palm shall not be considered caked.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

4.5.7.2 Sulfur dioxide content. The sulfur dioxide content of the product shall be determined in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Food Additives: Direct, Section: Total Sulfurous Acid (with FDA modifications).

4.5.7.3 Moisture. Test for moisture content of the product shall be performed as follows: Weigh five 3 to 4-gram samples in aluminum weighing dishes (approximately 2 to 2-1/2 inches in diameter and 3/4 inch in depth) with tight-fitting covers. Place the dishes with cocked lids in a vacuum oven and dry for 30 hours at 60°C under a pressure of not more than 100 millimeters of mercury. During drying, admit to the oven a slow current of air (approximately two bubbles per second) dried by passing through concentrated sulfuric acid. Remove the dishes and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples.

4.5.7.4 pH. pH shall be determined in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Beverages: Malt Beverages and Brewing Materials, Method: Hydrogen Ion Concentration (pH) Potentiometric Method.

4.5.7.5 Degrees Brix. Degrees Brix corrected to 20°C shall be determined by means of a refractometer, either directly with the use of a sugar refractometer or correlated with refractive indices of sucrose solutions at 20°C, if the readings are in terms of refractive index. No correction shall be made for insoluble solids or acid content. A sample of a thoroughly mixed product, rehydrated according to footnote 4/ of table I, shall be comminuted before determining the degrees Brix. If it is difficult to obtain a clear reading, the applesauce can be filtered through a milk-filter pad.

4.5.8 Can condition examination. Examination of filled and sealed cans shall be in accordance with the U.S. Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.5.9 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: Required markings missing, incorrect, or illegible;
Reinforced with other than nonmetallic strapping or tape (except for type II, class F loads);
Dimensions of pads not as specified;
Interior or packing with fiberboard pads not as specified.

Minor: Arrangement or number of cans not as specified.

4.5.10 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A. A net weight of 22 ounces of product shall be filled into a size 401 by 411, open-top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than 0.25-pound per base box commercial electrolytic tin plate (see 4.5.1.2), and shall be coated outside with a coating conforming to type I or type II of TT-C-495. A minus 1/2 ounce tolerance will be allowed in any one can provided the average net weight of the cans, inspected in accordance with 4.5.3, is not less than the stated net weight. The product shall be gas-packed by first removing the air from the filled can and replacing it with nitrogen. When the ends of the can are clinched prior to vacuumizing and gassing, the body flange or lid should be indented or dimpled to prevent seal-off trouble during the nitrogen flush. If tip soldering is used to close the exhaust vents for nitrogen filling, a suitable noncorrosive flux shall be used. The instant applesauce shall be unit packed under nitrogen so that the oxygen content of the gases in the filled and sealed container, not sooner than 2 days after filling, shall not exceed 2.0 percent when tested in accordance with 4.5.5. The filled and sealed can shall not show leakage when examined in accordance with 4.5.6. Alternatively, the cans shall be gas-packed as follows:

The product shall be gas-packed by first removing the air from the filled cans for not less than 30 seconds at not less than 28 inches of vacuum. While the cans are still under vacuum, they shall be flushed with nitrogen and brought to at least 3 psi of pressure for not less than 30 seconds and sealed. When the ends of the cans are clenched prior to vacuumizing and gassing, the body flange or lid should be indented or dimpled to prevent seal-off trouble during the nitrogen flush.

5.1.2 Level C. The product shall be preserved in accordance with 5.1.1, except that cans with or without commercial exterior coating will be acceptable.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Twenty-four cans of product, arranged four in length, three in width, and two in depth shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. The tiered cans shall be separated with a full length and width fiberboard pad made of the same material as the box. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of unit load specified (see 6.1). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

5.2.2 Level B packing. Twenty-four cans of product arranged four in length, three in width, and two in depth shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V3c, V3s, or V4s of PPP-B-636. The tiered cans shall be separated with a full length and width fiberboard pad made of the same material as the box. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Twenty-four cans of product arranged four in length, three in width, and two in depth shall be packed in a snug-fitting shipping container. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in accordance with the appendix of PPP-B-636.

5.3 Unitized loads. When specified, the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497, and as follows:

APPLESAUCE, INSTANT

Preserved with sulphur dioxide

GAS-PACKED

DIRECTIONS FOR USE

Applesauce, instant 1/	22 ounces (1 No. 2-1/2 can)
Water, cool	3 quarts

Place contents in suitable container.
Stir to break up lumps.
Add half the water. Stir until smooth.
Stir in remaining water.
Allow to stand 10 minutes before serving.

1/ Caked applesauce can be prepared for reconstitution by softening in a 200 °F oven and breaking apart by hand.

Yield: 50 two-ounce (1/4 cup) servings

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.5.3 and 6.2).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.33A/DSAR 4145.7.

6.4 Subject term (key word) listing.

Applesauce
Canned food
Rations

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8915-0828

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-35045D	Applesauce, Instant

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:51Z
Check the source to verify that this is the current version before use.